

Common Spatial Patterns (CSP)

1) Objectives of CSP

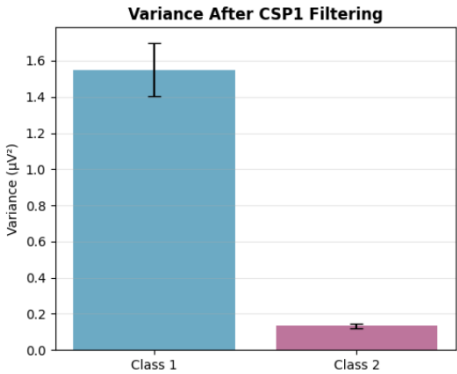
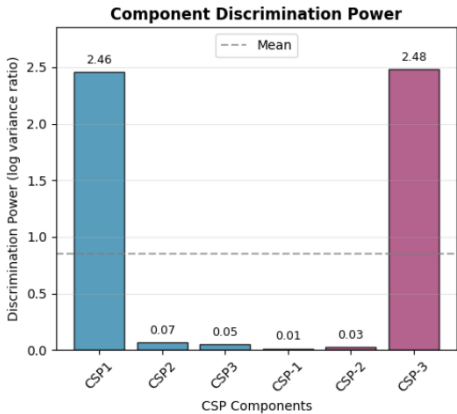
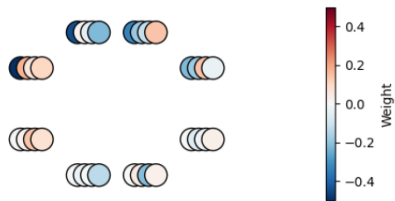
- Find optimal spatial filters that maximize variance for one class while minimizing it for another
- Discriminate between two mental states (e.g., left vs right hand motor imagery)
- Extract spatial patterns that maximize separability of EEG signals from different conditions
- Reduce dimensionality while preserving class-discriminative information.

2) How It works

- CSP is a spatial filtering technique that:
- Computes **covariance matrices** for two classes of EEG data (e.g., class A and class B)
- Solves a **generalized eigenvalue problem** to find filters that maximize variance ratio between classes
- Selects **top and bottom components** (e.g., CSP1, CSP2, CSP-1, CSP-2)
- Projects EEG data onto these spatial filters to enhance class separation
- Resulting features are optimal for classification (e.g., in Brain-Computer Interfaces)
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Common Spatial Patterns (CSP) Analysis

Spatial Patterns (Top 4 CSP Components)



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CSP Analysis Summary
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Number of channels: 8
Trials per class: 50
CSP components extracted: 6

Discrimination Power by component:
CSP1: 2.460
CSP2: 0.065
CSP3: 0.050
CSP-1: 0.014
CSP-2: 0.026
CSP-3: 2.478

Best discriminating component: CSP-3
Mean discrimination power: 0.849

